$\boxed{\text{Valid conclusion:}}$ 1) Last Monday was either Sunny or partly sunny or Last Friday was Sunny 2) I did not work on Tuesday or it was partly sunny on Tuesday.

c) "All insects have six legs." "Dragonflies are insects." "Spiders do not have six legs." "Spiders eat dragonflies."

$\boxed{\text{Soln:}}$ $I(x)$: x is an insect. $L(x)$: x has six legs $D(x)$: x is a dragonfly $E(x,y)$: x eats y

$S(x)$: x is a Spider $\boxed{\text{Domain:}}$ Set of all living beings

$\boxed{\text{Premise:}}$ $\forall x(I(x) \Rightarrow L(x))$ ① ② $\forall x(D(x) \Rightarrow I(x))$ ③ $\forall x(S(x) \Rightarrow \neg L(x))$

④ $\forall x \forall y(S(x) \wedge D(y) \Rightarrow E(x,y))$

Steps	Reason
1. $\forall x(I(x) \Rightarrow L(x))$	Premise
2. $\forall x(D(x) \Rightarrow I(x))$	Premise
3. $\forall x(S(x) \Rightarrow \neg L(x))$	Premise
4. $\forall x \forall y(S(x) \wedge D(y) \Rightarrow E(x,y))$	Premise
5. $I(c) \Rightarrow L(c)$	Universal instantiation on 1
6. $D(c) \Rightarrow I(c)$	Universal instantiation on 2
7. $D(c) \Rightarrow L(c)$	Hypothetical syllogism on 6,5
8. $S(c) \Rightarrow \neg L(c)$	Universal instantiation on 3
9. $\neg L(c) \Rightarrow \neg I(c)$	Contrapositive of 5
10. $S(c) \Rightarrow \neg I(c)$	Hypothetical syllogism on 8,9
11. $\forall x(D(x) \Rightarrow L(x))$	Universal generalization on 7
12. $\forall x(S(x) \Rightarrow \neg I(x))$	Universal generalization on 10

$\boxed{\text{Valid conclusion:}}$ 1) Dragonflies have six legs 2) Spiders are not insects